

Raymond Tsai

Seattle, Washington

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EDUCATION

University of Washington

Seattle, WA

Bachelor of Science in Computer Science and Software Engineering, Minor in Mathematics Sept. 2023 – Present

- Relevant Coursework: Data Structures & Discrete Math, Hardware & Computer Organization, Operating Systems, Database Systems, Network Design & Programming, Cloud Computing, Embedded Systems, Computer Vision

EXPERIENCE

Platform Engineer Intern

June 2026 – September 2026

Costco IT

Issaquah, WA

- Engineered an Azure-hosted ETL pipeline in Python that consolidates DocuSign usage metadata for 7,000 enterprise users into a centralized enterprise SQL database, reducing report query times by 79%.
- Containerized the pipeline as a cron-scheduled Azure Container Apps job with credentials secured in Azure Key Vault, running batched incremental loads with retry logic and idempotent writes.
- Provisioned Azure infrastructure as code with Terraform and automated build, test, and deployment through a GitHub Actions CI/CD pipeline, standardizing container releases across environments.

CSE Intern

January 2026 – June 2026

AI4DeafBlind, Helen Keller Foundation

Seattle, WA

- Created a real time Speech to Braille demo in Python using whisper.cpp and pybrl for deaf and blind individuals
- Developing real time text to Braille streaming to Brailiant display to prepare for future smart glass technology

Hackathons

UWBHacks '24, UWBHacks '25, DubHacks '25, UWBHacks '26 (Finalist)

PROJECTS

GridLock (UWBHacks '26 Finalist) | *React.js, FastAPI, Docker* | <http://178.128.71.114>

April 2026

- Designed interactive 3D map of Seattle with MapLibre that renders 3D buildings, transit routes, and a heatmap
- Built a FastAPI backend that serves Server-Sent Events for live heatmap data, and simulation state switching
- Applied geospatial joins, distance-based weighting, GTFS parsing, time-binned feature engineering, and gradient-boosted regression to build a city-agnostic transit demand simulation ML model and pipeline
- Dockerized the app with Compose, split frontend and backend into separate images, implemented CI/CD pipeline via GitHub Actions to perform automated tests, and attached Caddy & Nginx reverse proxy for scalability

Schedulux | *React.js, Express.js, PostgreSQL, AWS* | <https://schedulux.net>

May 2025 - Ongoing

- Developed a React 18/TypeScript frontend using TailwindCSS, TanStack Query, and Zustand that allows small businesses to advertise small, irregular time slots to potential customers and display example vendor work
- Implemented Node.js backend with PostgreSQL database to guarantee data integrity and detect real-time conflicts
- Deployed container to AWS EC2 and RDS via Docker Compose and Caddy reverse proxy for automated HTTP

Palorant | *C++/C, RayLib, OpenGL*

December 2025

- Developed a 3D multiplayer FPS engine in C++ and Raylib, implementing custom physics, collision detection, and raycasting mechanics from scratch, allowing for a smooth multiplayer gameplay experience.
- Engineered a hybrid UDP/TCP networking architecture with custom POSIX sockets to synchronize high-frequency player state and reliable chat messages across the client-server topology.
- Built a server system handling lobby creation, hitcan validation, and game state replication for multiple clients.

TECHNICAL SKILLS

Coding Languages: Go, Java, Python, C/C++/C#, JavaScript/TypeScript, Terraform, Rust, HTML/CSS, SQL
Libraries/Frameworks: Flutter, Next.js, React.js, Express.js, Node.js, Flask, PostgreSQL, MySQL, Tailwind CSS
Cloud: Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP)
Developer Tools: Jira, Git, Docker, Kubernetes, Caddy, Nginx, Supabase, Prisma ORM
Languages: English, Mandarin Chinese, Japanese